



Mansoura University
Faculty of Computers and Information
Department of Information System



[IS422P] Data Mining

Grade:4 Th

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MINING FREQUENT PATTERNS, ASSOCIATIONS, & CORRELATIONS



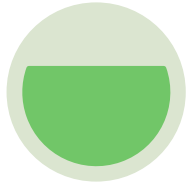


The Basics

Market Basket
Analysis

Frequent Itemsets

Association Rules

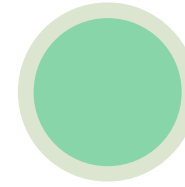


Frequent Itemset Mining Methods

Apriori Algorithm

Generating
Association Rules
from Frequent
Itemsets

FP-Growth



Pattern Evaluation Methods



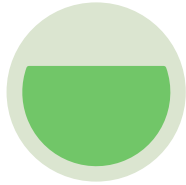


The Basics

Market Basket
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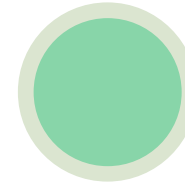


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Mining Methods

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Pattern Evaluation
Methods



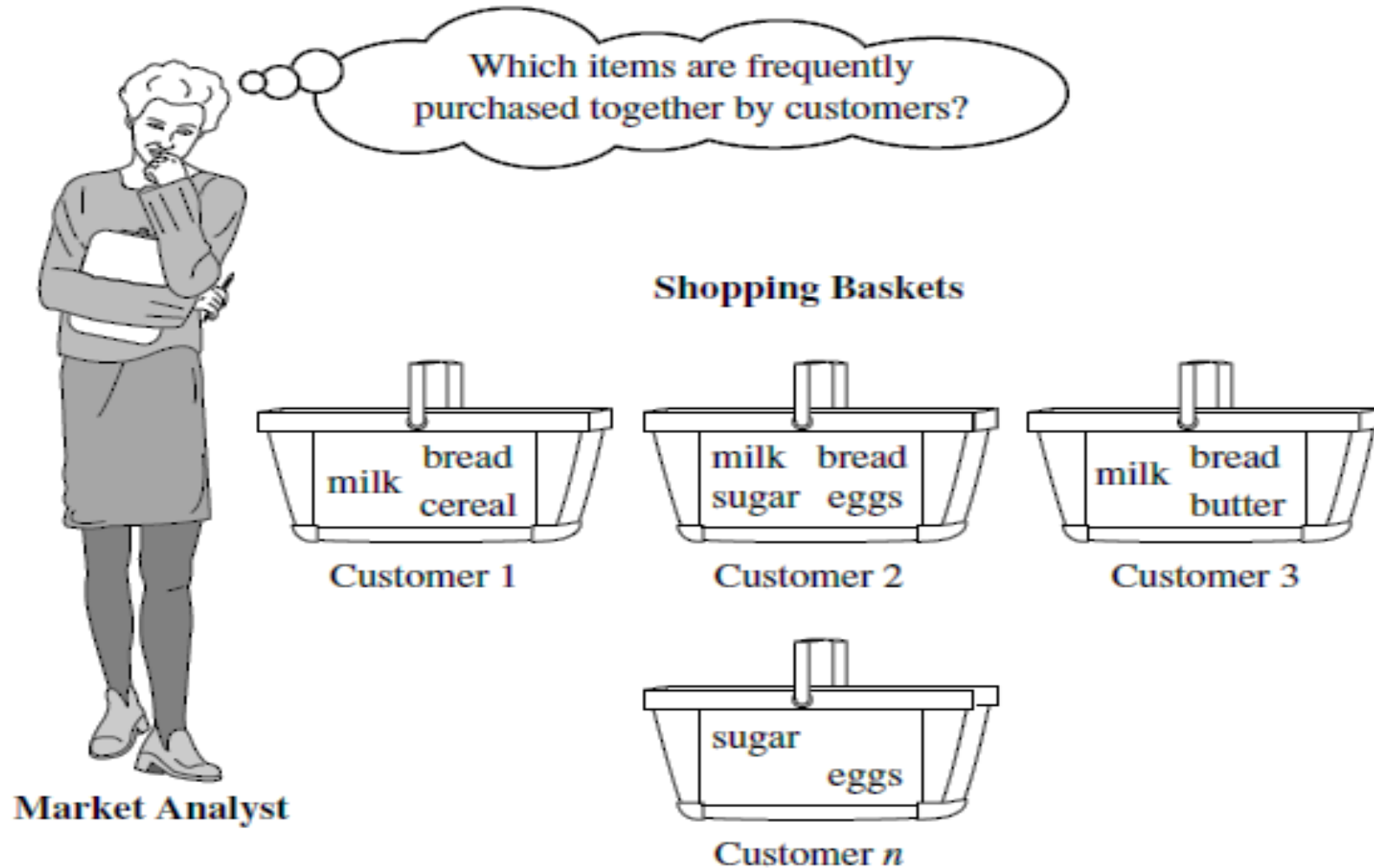
THE BASICS

WHAT IS FREQUENT PATTERN ANALYSIS?

- Frequent pattern: a pattern (a set of items, subsequences, substructures, etc.) that occurs frequently in a data set
- First proposed by Agrawal, Imielinski, and Swami [AIS93] in the context of frequent itemsets and association rule mining



THE BASICS



THE BASICS

- **Motivation:** Finding inherent regularities in data
 - What products were often purchased together?— Beer and diapers?!
 - What are the subsequent purchases after buying a PC?
 - What kinds of DNA are sensitive to this new drug?
 - Can we automatically classify web documents?
- **Applications**
 - Basket data analysis, cross-marketing, catalog design, sale campaign analysis, Web log (click stream) analysis, and DNA sequence analysis



THE BASICS

- Frequent Pattern are itemsets that appear frequently in a data set (e.g.Transaction record)
- Items that are frequently associated (e.g purchased) together can be represented as association rules

Computer → antivirus_SW [Support = 2% , Confidence =60%]

- **Support** and **Confidence** are measures of rule interestingness
- 2% Support means 2% of Transactions Show that computers and antivirus_SW are bought Together
- 60% Confidence means 60 % of customers who bought a computer also bought antivirus_SW



THE BASICS

FREQUENT ITEM-SETS

- Itemset $X = \{x_1, \dots, x_k\}$ ex: $X = \{A, B, C, D, E, F\}$
- Find all the rules $X \rightarrow Y$ with minimum support and confidence
 - support, s , probability that a transaction contains $X \cup Y$
 - confidence, c , conditional probability that a transaction having X also contains Y

$$\text{support } X \rightarrow Y = P(X \cup Y) = \frac{n(X \cup Y)}{N}$$

$$\text{confidence } (X \rightarrow Y) = P(Y|X) = \frac{n(X \cup Y)}{n(X)}$$



THE BASICS

ASSOCIATION RULES

Ex: Let $\text{min_Sup.} = 50\%$, $\text{min_conf.} = 50\%$

Frequent Patterns:

$\{A:3, B:3, D:4, E:3, AD:3\}$

Association rules:

$A \rightarrow D$ (60%, 100%)

$D \rightarrow A$ (60%, 75%)

$$\text{conf}(A \rightarrow D) = \frac{3}{3} = 100\%$$

$$\text{conf}(D \rightarrow A) = \frac{3}{4} = 75\%$$

Transaction-id	Items bought
10	A, B, D
20	A, C, D
30	A, D, E
40	B, E, F
50	B, C, D, E, F

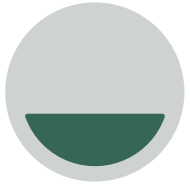


THE BASICS

ASSOCIATION RULES

- If frequency of itemset I satisfies min_support count then I is a frequent itemset
- If a rule satisfies min_support and min_confidence thresholds, it is said to be strong
 - problem of mining association rules reduced to mining frequent itemsets
- Association rules mining becomes a two-step process:
 - Find all frequent itemsets that occur at least as frequently as a predetermined min_support count
 - Generate strong association rules from the frequent itemsets that satisfy min_support and min_confidence



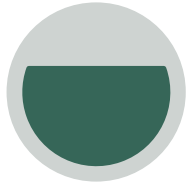


The Basics

Market Basket
Analysis

Frequent Itemsets

Association Rules

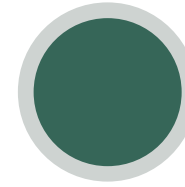


Frequent Itemset Mining Methods

Apriori Algorithm

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FP-Growth



Pattern Evaluation Methods



MINING FREQUENT ITEMSETS

APRIORI ALGORITHM

- Goes as follows:
 - Find frequent 1-itemsets $\rightarrow L_1$
 - Use L_1 to find frequent 2-itemsets $\rightarrow L_2$
 - ... until no more frequent k-itemsets can be found
- Each L_k itemset requires a full dataset scan
- To improve efficiency, use the Apriori property:
 - “All nonempty subsets of a frequent itemset must also be frequent” – if a set cannot pass a test, all of its supersets will fail the same test as well – if $P(I) < \text{min_support}$ then $P(I \cup A) < \text{min_support}$



MINING FREQUENT ITEMSETS

APRIORI ALGORITHM

Transactional data example
N=10, min_supp count=2

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

Scan dataset for
count of each
candidate

C_1

Itemset	Support count
{I1}	7
{I2}	8
{I3}	6
{I4}	2
{I5}	2

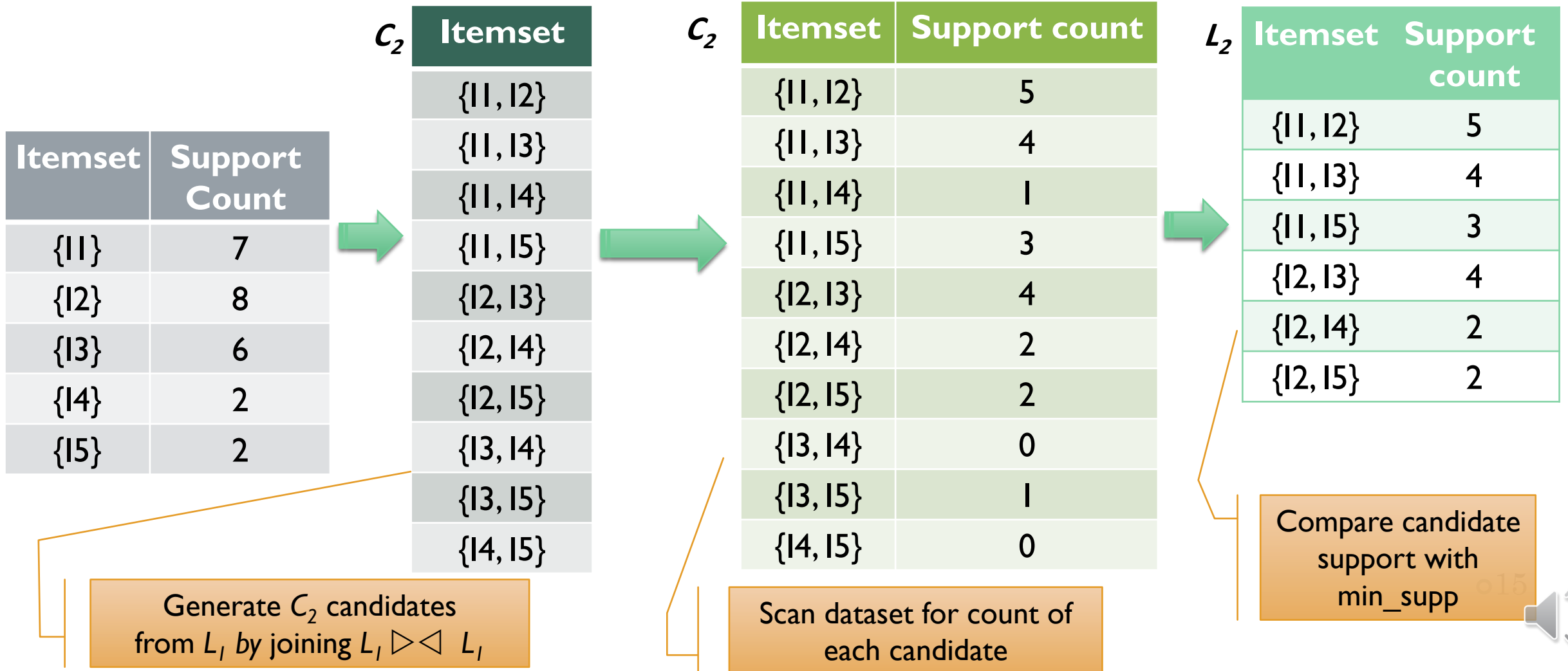
Compare candidate
support with
min_support

L_1

Itemset	Support Count
{I1}	7
{I2}	8
{I3}	6
{I4}	2
{I5}	2

MINING FREQUENT ITEMSETS

APRIORI ALGORITHM



MINING FREQUENT ITEMSETS

APRIORI ALGORITHM

$C_3 = L_2 \triangleright \triangleleft L_2 = \{\{I1, I2, I3\}, \{I1, I2, I5\},$
 ~~$\{I1, I3, I5\}, \{I2, I3, I4\}, \{I2, I3, I5\}, \{I2, I4, I5\}\}$~~

Not all subsets are frequent $\rightarrow$ **Prune** (Apriori property)

Itemset	Support count
{I1, I2}	5
{I1, I3}	4
{I1, I5}	3
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2



Itemset
{I1, I2, I3}
{I1, I2, I5}



C_3	
Itemset	Support count
{I1, I2, I3}	2
{I1, I2, I5}	2

Scan dataset for count of each candidate



L_3	
Itemset	Support count
{I1, I2, I3}	2
{I1, I2, I5}	2

Compare candidate support with min_supp

Generate C_3 candidates from L_2 by **joining** $L_2 \triangleright \triangleleft L_2$



Two joining (lexicographically ordered) k-itemsets must share first k-1 items $\rightarrow \{I1, I2\}$ is not joined with $\{I2, I4\}$



MINING FREQUENT ITEMSETS

APRIORI ALGORITHM

Itemset	Support count
{11, 12, 13}	2
{11, 12, 15}	2



Itemset
{11, 12, 13, 15}



Not all subsets are frequent → **Pruning**

$C_4 = \phi \rightarrow$ Terminate



APRIORI ALGORITHM

Algorithm: Apriori. Find frequent itemsets using an iterative level-wise approach based on candidate generation.

Input:

- D , a database of transactions;
- min_sup , the minimum support count threshold.

Output: L , frequent itemsets in D .

Method:

Generate C_k using L_{k-1} to find L_k

```
(1)  $L_1 = \text{find\_frequent\_1-itemsets}(D)$ ;  
(2) for ( $k = 2; L_{k-1} \neq \phi; k++$ ) {  
(3)    $C_k = \text{apriori\_gen}(L_{k-1})$ ;  
(4)   for each transaction  $t \in D$  { // scan  $D$  for counts  
(5)      $C_t = \text{subset}(C_k, t)$ ; // get the subsets of  $t$  that are candidates  
(6)     for each candidate  $c \in C_t$   
(7)        $c.\text{count}++$ ;  
(8)   }  
(9)    $L_k = \{c \in C_k | c.\text{count} \geq min\_sup\}$   
(10) }  
(11) return  $L = \cup_k L_k$ ;
```

Join

```
procedure  $\text{apriori\_gen}(L_{k-1}:\text{frequent } (k-1)\text{-itemsets})$   
(1) for each itemset  $l_1 \in L_{k-1}$   
(2)   for each itemset  $l_2 \in L_{k-1}$   
(3)     if ( $(l_1[1] = l_2[1]) \wedge (l_1[2] = l_2[2])$   
        $\wedge \dots \wedge (l_1[k-2] = l_2[k-2]) \wedge (l_1[k-1] < l_2[k-1])$ ) then {  
(4)        $c = l_1 \bowtie l_2$ ; // join step: generate candidates  
(5)       if  $\text{has\_infrequent\_subset}(c, L_{k-1})$  then  
(6)         delete  $c$ ; // prune step: remove unfruitful candidate  
(7)       else add  $c$  to  $C_k$ ;  
(8)     }  
(9) return  $C_k$ ;
```

Prune

```
procedure  $\text{has\_infrequent\_subset}(c:\text{candidate } k\text{-itemset};$   
        $L_{k-1}:\text{frequent } (k-1)\text{-itemsets})$ ; // use prior knowledge  
(1) for each  $(k-1)$ -subset  $s$  of  $c$   
(2)   if  $s \notin L_{k-1}$  then  
(3)     return TRUE;  
(4) return FALSE;
```



MINING FREQUENT ITEMSETS

GENERATING ASSOCIATION RULES FROM FREQUENT ITEMSETS

- Association rules can be generated using the confidence equation, as follows”

$$\text{confidence}(A \Rightarrow B) = P(B|A) = \frac{\text{support_count}(A \cup B)}{\text{support_count}(A)}$$

- For each frequent itemset L, generate all nonempty subset of L
- For every nonempty subset S of L, output rule: $S \rightarrow L-S$

- $\frac{\text{Supportcount}(L)}{\text{Supportcount}(S)} \geq \text{min_conf},$

where min_conf is the minimum confidence threshold.



MINING FREQUENT ITEMSETS

GENERATING ASSOCIATION RULES FROM FREQUENT ITEMSETS

Itemset	Support count
{11, 12, 13}	2
{11, 12, 15}	2

Nonempty subsets
{11, 12}
{11, 15}
{12, 15}
{11}
{12}
{15}

Association Rules
{11, 12} → 15
{11, 15} → 12
{12, 15} → 11
11 → {12, 15}
12 → {11, 15}
15 → {11, 12}

Confidence
$2/5 = 40\%$
$2/2 = 100\%$
$2/2 = 100\%$
$2/7 = 28\%$
$2/8 = 25\%$
$2/2 = 100\%$

For a *min_confidence* = 70%



MINING FREQUENT ITEMSETS

FP-GROWTH

- To avoid costly candidate generation
- Divide-and-conquer strategy:
- Compress database representing frequent items into a frequent pattern tree (FP-tree) – 2 passes over dataset
- Divide compressed database (FP-tree) into conditional databases, then mine each for frequent itemsets – traverse through the FP-tree



MINING FREQUENT ITEMSETS

FP-GROWTH

Transactional data example
N=10, min_supp count=2

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

Scan dataset for count
of each candidate

C_1

Itemset	Support count
{I1}	7
{I2}	8
{I3}	6
{I4}	2
{I5}	2

Compare candidate
support with min_supp

L_1 - Reordered

Itemset	Support count
{I2}	8
{I1}	7
{I3}	6
{I4}	2
{I5}	2



MINING FREQUENT ITEMSETS

FP-GROWTH – FP-TREE CONSTRUCTION

L_1 - Reordered

Itemset	Support count
{I2}	8
{I1}	7
{I3}	6
{I4}	2
{I5}	2

FP-tree



TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
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T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

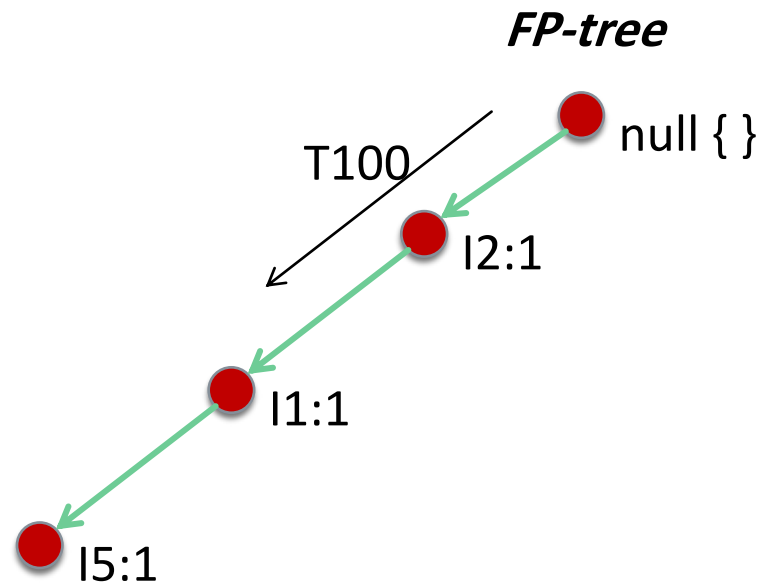


MINING FREQUENT ITEMSETS

FP-GROWTH – FP-TREE CONSTRUCTION

L_1 - Reordered

Itemset	Support count
{I2}	8
{I1}	7
{I3}	6
{I4}	2
{I5}	2



Order of items is kept throughout path construction, with common prefixes shared whenever applicable

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

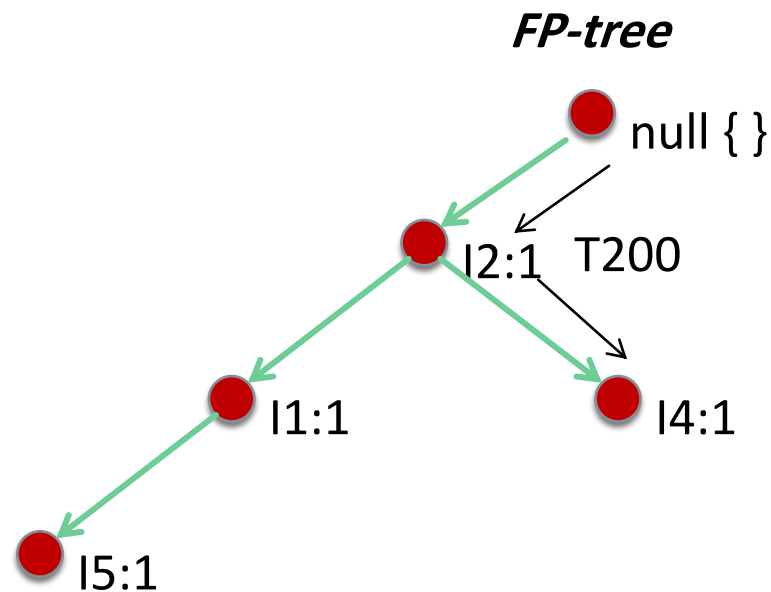


MINING FREQUENT ITEMSETS

FP-GROWTH – FP-TREE CONSTRUCTION

L_1 - Reordered

Itemset	Support count
{I2}	8
{I1}	7
{I3}	6
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{I5}	2



TID	List of items
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T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

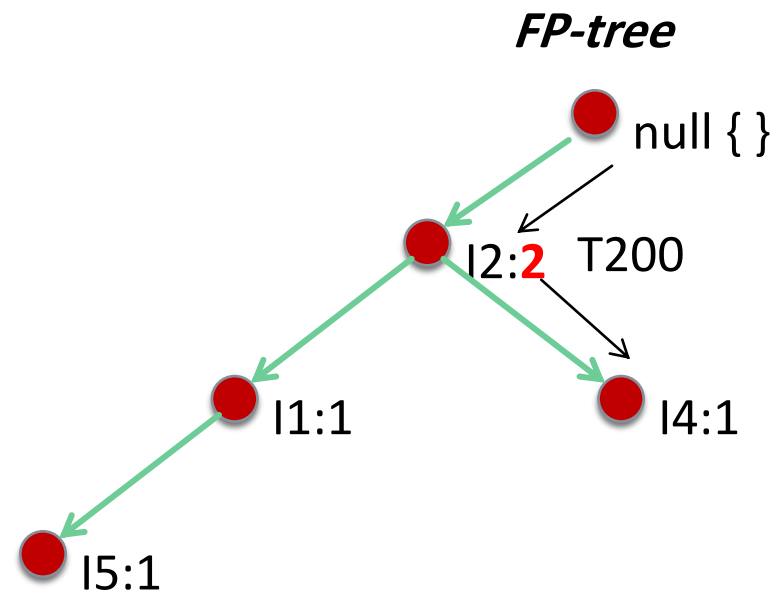


MINING FREQUENT ITEMSETS

FP-GROWTH – FP-TREE CONSTRUCTION

L_1 - Reordered

Itemset	Support count
{I2}	8
{I1}	7
{I3}	6
{I4}	2
{I5}	2



TID	List of items
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T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

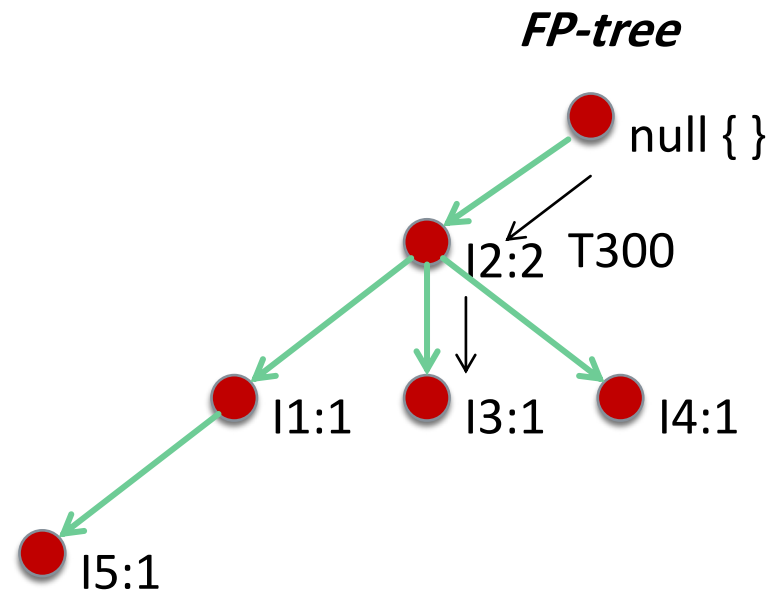


MINING FREQUENT ITEMSETS

FP-GROWTH – FP-TREE CONSTRUCTION

L_1 - Reordered

Itemset	Support count
{I2}	8
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{I5}	2



TID	List of items
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T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

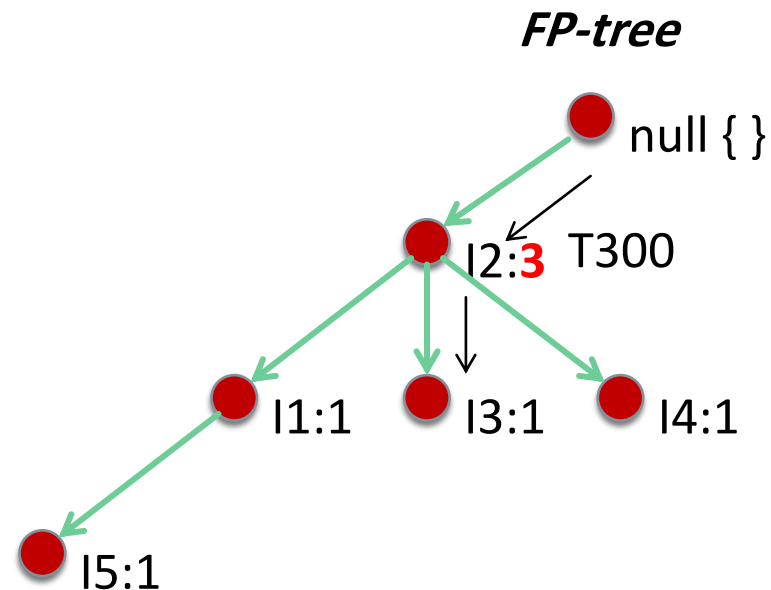


MINING FREQUENT ITEMSETS

FP-GROWTH – FP-TREE CONSTRUCTION

L_1 - Reordered

Itemset	Support count
{I2}	8
{I1}	7
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{I5}	2



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T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2



MINING FREQUENT ITEMSETS

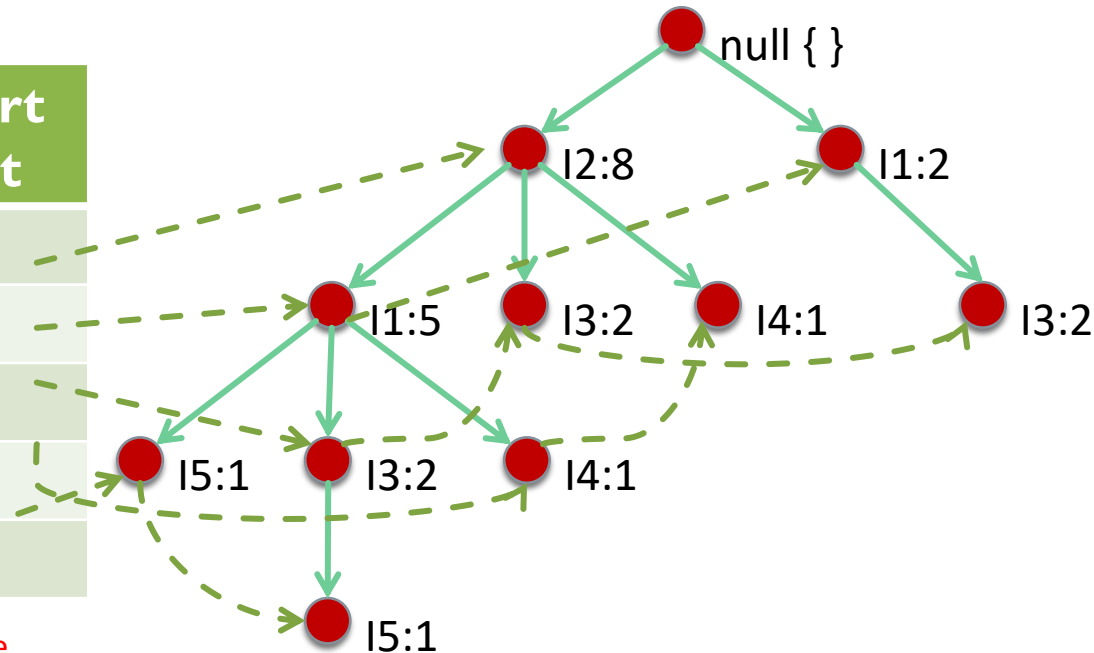
FP-GROWTH – FP-TREE CONSTRUCTION

L_1 - Reordered

Itemset	Support count
{I2}	8
{I1}	7
{I3}	6
{I4}	2
{I5}	2

For Tree Traversal

FP-tree



TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

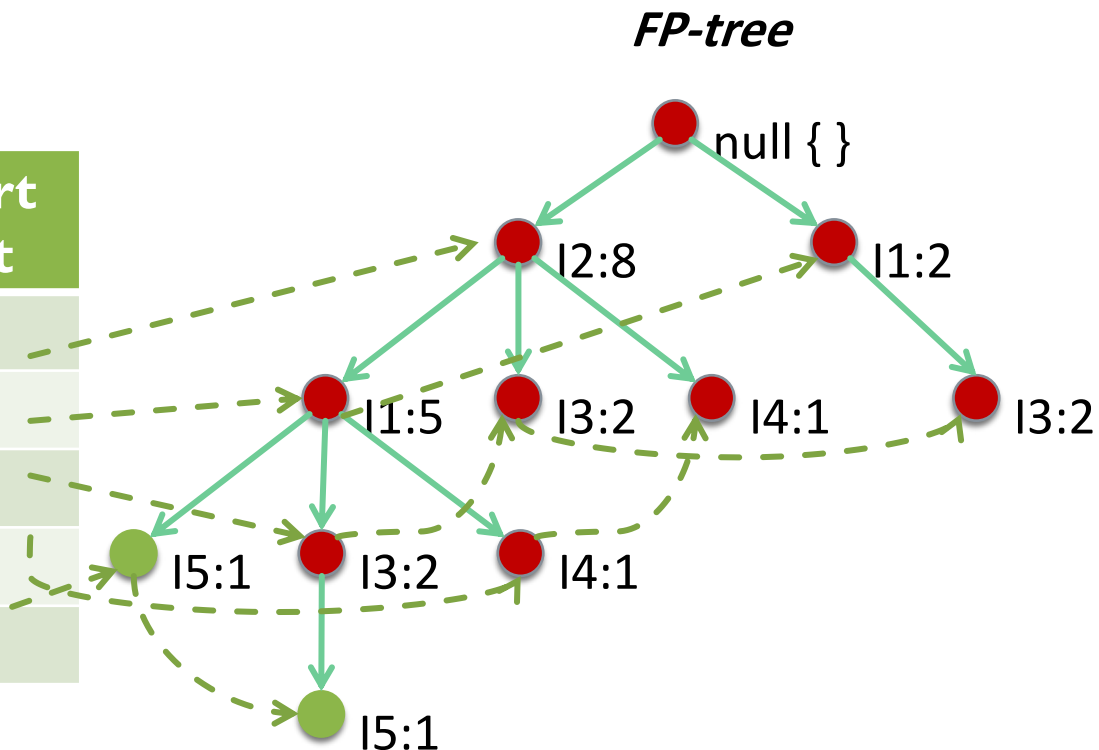


MINING FREQUENT ITEMSETS

FP-GROWTH – FREQUENT PATTERNS MINING

L_1 - Reordered

Itemset	Support count
{I2}	8
{I1}	7
{I3}	6
{I4}	2
{I5}	2



Bottom-up algorithm – start from leaves and go up to root

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2



MINING FREQUENT ITEMSETS

FP-GROWTH – CONDITIONAL FP-TREE CONSTRUCTION

For I5

L₁ - Reordered

Itemset	Support count	Linked link
{I2}	8	
{I1}	7	
{I3}	6	
{I4}	2	
{I5}	2	

FP-tree

 null { }

Eliminate I5

Eliminate transactions
not including I5

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2



MINING FREQUENT ITEMSETS

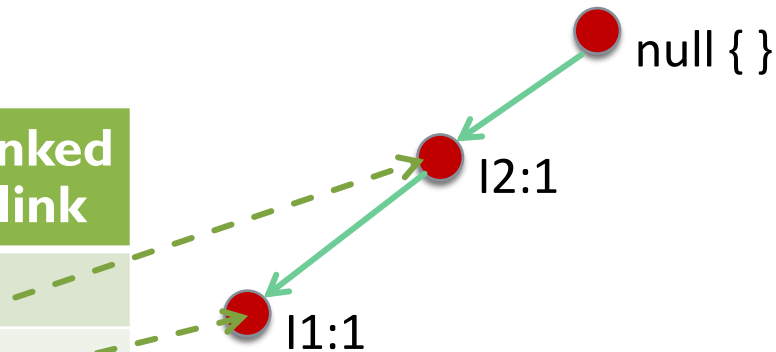
FP-GROWTH – CONDITIONAL FP-TREE CONSTRUCTION

For I5

L_1 - Reordered

Itemset	Support count	Linked link
{I2}	8	
{I1}	7	
{I3}	6	
{I4}	2	
{I5}	2	

FP-tree



Eliminate I5

Eliminate transactions
not including I5

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2



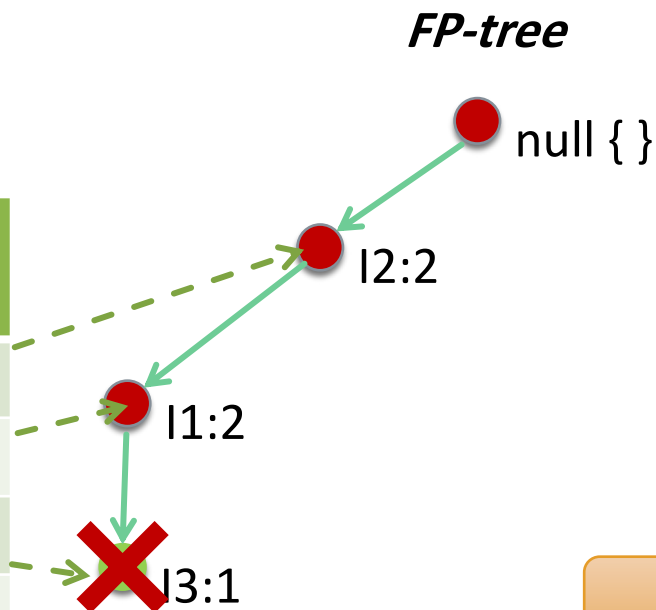
MINING FREQUENT ITEMSETS

FP-GROWTH – CONDITIONAL FP-TREE CONSTRUCTION

For I5

L_1 - Reordered

Itemset	Support count	Linked link
{I2}	8	
{I1}	7	
{I3}	6	
{I4}	2	
{I5}	2	



Eliminate I5

Eliminate transactions
not including I5

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2



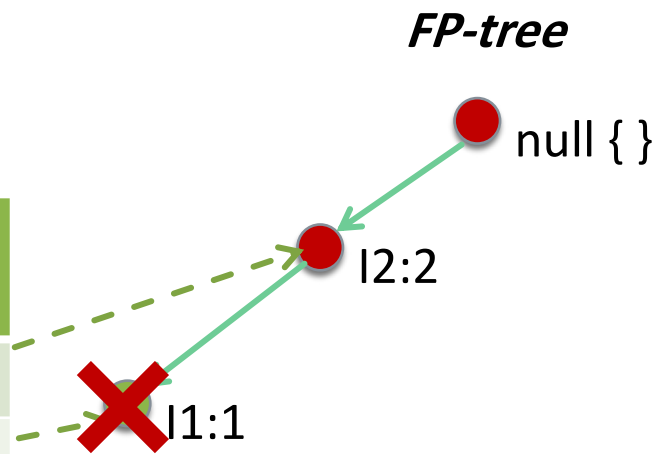
MINING FREQUENT ITEMSETS

FP-GROWTH – CONDITIONAL FP-TREE CONSTRUCTION

For I4

L_1 - Reordered

Itemset	Support count	Linked link
{I2}	8	
{I1}	7	
{I3}	6	
{I4}	2	
{I5}	2	



Eliminate I4

Eliminate transactions
not including I4

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2



MINING FREQUENT ITEMSETS

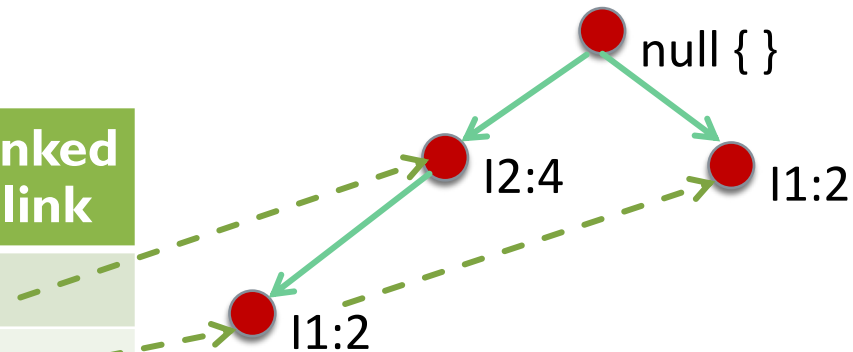
FP-GROWTH – CONDITIONAL FP-TREE CONSTRUCTION

For I3

L_1 - Reordered

Itemset	Support count	Linked link
{I2}	8	
{I1}	7	
{I3}	6	
{I4}	2	
{I5}	2	

FP-tree



Eliminate I3

Eliminate transactions
not including I3

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3 , I5
T900	I1, I2, I3
T1000	I1, I2

MINING FREQUENT ITEMSETS

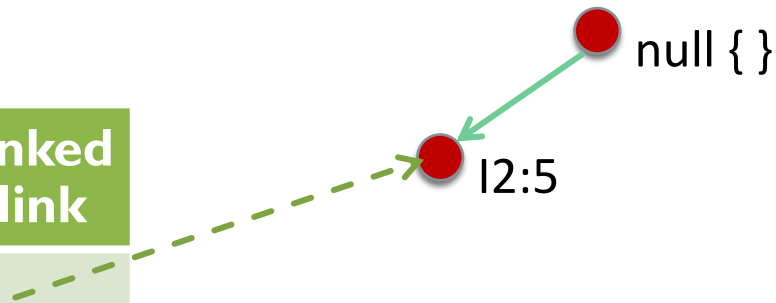
FP-GROWTH – CONDITIONAL FP-TREE CONSTRUCTION

For I1

L_1 - Reordered

Itemset	Support count	Linked link
{I2}	8	
{I1}	7	
{I3}	6	
{I4}	2	
{I5}	2	

FP-tree



Eliminate I1

Eliminate transactions
not including I1

TID	List of items
T100	I1 , I2, I5
T200	I2, I4
T300	I2, I3
T400	I1 , I2, I4
T500	I1 , I3
T600	I2, I3
T700	I1 , I3
T800	I1 , I2, I3, I5
T900	I1 , I2, I3
T1000	I1 , I2

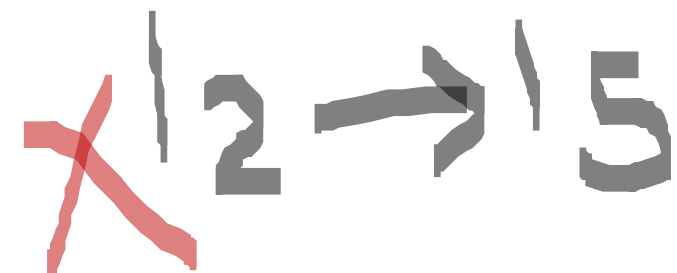
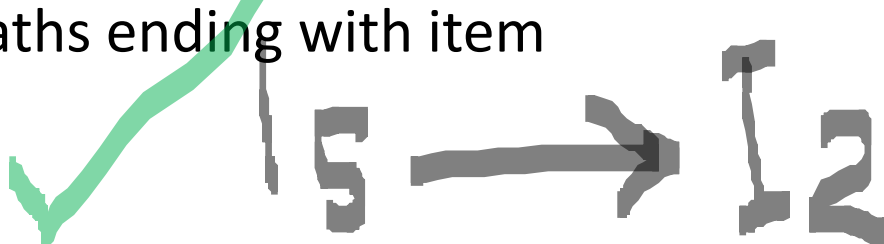


MINING FREQUENT ITEMSETS

FP-GROWTH

Item	Conditional Pattern Base	Conditional FP-tree	Frequent Patterns Generated
15	$\{\{I2, I1: 1\}, \{I2, I1, I3: 1\}\}$	$\langle I2:2, I1:2 \rangle$	$\{I2, I5: 2\}, \{I1, I5: 2\}, \{I2, I1, I5: 2\}$
14	$\{\{I2, I1: 1\}, \{I2: 1\}\}$	$\langle I2:2 \rangle$	$\{I2, I4: 2\}$
13	$\{\{I2, I1: 2\}, \{I2: 2\}, \{I1: 2\}\}$	$\langle I2:4, I1:2 \rangle, \langle I1:2 \rangle$	$\{I2, I3: 4\}, \{I1, I3: 4\}, \{I2, I1, I3: 2\}$
11	$\{\{I2: 5\}\}$	$\langle I2:5 \rangle$	$\{I2, I1: 5\}$

Paths ending with item



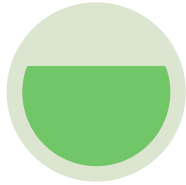


The Basics

Market Basket
Analysis

Frequent Itemsets

Association Rules

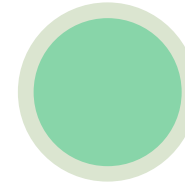


Frequent Itemset Mining Methods

Apriori Algorithm

Generating
Association Rules
from Frequent
Itemsets

FP-Growth



Pattern Evaluation Methods



PATTERN EVALUATION METHODS

- Not all association rules are interesting
 - $\text{Buys}(X, \text{"Computer games"}) \rightarrow \text{buys}(X, \text{"Videos"})$ [40%, 66%]
 - $P(\text{"videos"})$ is 75% > 66%
 - The two items are negatively associated means buying one decreases the likelihood of buying the other
 - We need to measure “real strength” of rule
- **Correlation analysis**
 - $A \rightarrow B$ [support , confidence , correlation]



PATTERN EVALUATION METHODS

1. **Lift** = $\frac{P(A \cup B)}{P(A)P(B)}$
 - A and B are independent if $P(A \cup B) = P(A)P(B)$
 - Otherwise, **dependent and correlated** occurrence
 - If lift < 1, A is **Negatively correlated** with B
 - If lift > 1, A is **Positively correlated** with B A's occurrence "lifts" the occurrence of B
2. χ^2 → already discussed in previous lecture



QUESTIONS?

NEXT ...

